

Harish Natarajan Ravi

Software Engineer – Embedded, Perception and AI

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Professional Overview

Software Engineer with expertise in robotics, machine vision, and multi-sensor fusion for intelligent systems. Skilled in developing AI-driven perception and feature-based algorithms that enable robots to perceive, understand, and interact with environments. I aim to apply my strengths in software development, machine vision, & integration to build reliable robotics and automation solutions.

Primary stack: C/C++, Python, ROS2, Sensor Fusion, SLAM, Perception, Computer Vision, AI, ML/DNN, PyTorch, TensorFlow, Mapping, Tracking, LiDAR, Camera, Simulation, V&V, Robotics, Linux, Git, CI/CD

Work Experience

Research Assistant | Fraunhofer IVI | Part-time | Ingolstadt, DE **Nov 2025 – Present**

Tech stack: C++, Python, Sensor Fusion, ML/DL Frameworks, KF, Object Tracking, Calibration

- Currently developing 3D scene reconstruction, sensor fusion, and RGB-IR calibration pipelines.
- Researching AI-assisted multi-object tracking and state estimation for robust mobility perception.

Software Development Engineer | Mercor | Contract | Leonberg, DE **Jan 2025 – Oct 2025**

Tech stack: C++, Python, Linux, PyTorch, TensorFlow, ML/DL Frameworks

- Optimized embodied AI models for improved performance in autonomous systems.
- Integrated and validated AI components to enhance deployment efficiency and system reliability.

ADAS Research Engineer | Porsche | Internship | Mönsheim, DE **Sep 2023 – Sep 2024**

Tech stack: C++, Python, ROS, PCL, OpenCV, SLAM, Sensor Fusion, Git, Docker, CI/CD

- Built a complete SLAM-based parking map pipeline, improving localization accuracy.
- Boosted relocalisation robustness by integrating LiDAR and camera sensor fusion.
- Refined trajectory alignment by applying loop closure techniques to sensor data.
- Supported HMI development, image renderings, real-time simulation and on-vehicle testing.

System Development Engineer | Bosch | Part-time | Stuttgart, DE **Mar 2023 – Aug 2023**

Tech stack: C++, Python, SysML, MBSE, IBM Rhapsody and DOORS, ISO Standards, Jira

- Strengthened system clarity by modelling L3 perception modules and refined ISO requirements.
- Improved system validation coverage by executing structured, traceable verification workflows.

Research Assistant | Fraunhofer IEM | Part-time | Paderborn, DE

Sep 2021 – Feb 2023

Tech stack: Python, C++, MQTT, Eclipse HONO, Kubernetes, Grafana, Linux, Git

- Developed a car-to-cloud demonstrator to streamline F1 telemetry analysis.
- Contributed to an open-source SDV initiative by supporting integration with KUKSA stack.
- Enabled real-time data flow by implementing odometry and telemetry extraction modules.

Software Engineer | Robert Bosch | Full-time | Bengaluru, IN

Apr 2017 – Sep 2020

Tech stack: C/C++, Python, Perl, AUTOSAR, TDD, ASPICE, IBM DOORS

- Delivered production-grade service layers by developing diagnostics for global OEMs.
- Accelerated function validation by implementing and testing OBD/parking modules.
- Improved platform scalability by contributing to reusable diagnostic infrastructure.
- Ensured project stability by supporting release cycles with cross-functional teams.

Education

Master's in Computer Engineering | Paderborn University, DE

- Field of Study: Embedded Systems
- GPA: 1,8

Bachelor's in Electronics | Visvesvaraya Technological University, IN

- Field of Study: Electronics and Communication
- GPA: 2,3

Theses and Research Works

- Master Thesis - Generation and Relocalization of Parking Maps
- LiDAR and camera-based perception to enable object detection.
- Disaster Response Robots
- Autonomous Vehicle Perception
- Visual Odometry for Camera Motion Estimation
- Depth Estimation using Transformer models
- Object Classification and Detection using YOLO and RT-DETR models

Skills

Languages: English (Fluent), German (B1), Tamil (Native)

Driving License: German Class B | **Operating Systems:** Linux, Windows, WSL, QNX

Software, Tools and Platforms: ROS, CARLA, Machine and DNN Frameworks, Functional Testing, TRACE32, CAN, Data Handling and Analysis, Docker, CI/CD, SOCs, TDD, Confluence, Nvidia Jetson

Soft Skills: Effective Communication, Problem Solving, Critical Thinking, Teamwork, Adaptability.